

1 Methods

1.1 Virtual Element Method

In this work, the VEM is used to compute the displacement from the variational formulation of the momentum equation. For the sake of completeness, we give a brief introduction to the topic in the following section. The subsequent paragraph is strongly based on the excellent introduction given in [Beirão Da Veiga et al. \(2014\)](#). A theoretical overview can be found in [Beirão Da Veiga et al. \(2023\)](#), and a monograph with a more practical focus in [Wriggers et al. \(2024\)](#).

We will first explain the VEM for scalar valued problems in 2D. Afterward, we extend the explanation to 3D where the 2D techniques are reused for each face of the elements. Finally we show the application of the VEM for the [discretization](#) of the momentum equation.

As [an](#) introductory example, we use the well known Poisson problem. Find $u : \bar{\Omega} \rightarrow \mathbb{R}$ s.t.

$$-\Delta u = f \text{ in } \Omega, \quad u = 0 \text{ on } \partial\Omega. \quad (1)$$

Here, Ω denotes the open bounded problem domain and $\partial\Omega$ its boundary with $\bar{\Omega} = \Omega \cup \partial\Omega$. With

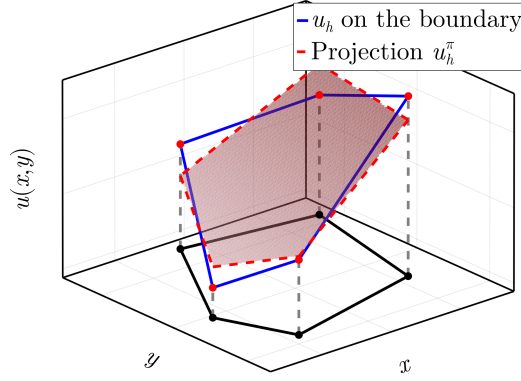
$$\begin{aligned} H_0^1(\Omega) &:= \{v \in H^1(\Omega) : v = 0 \text{ on } \partial\Omega\} \\ A(u, \delta u) &:= (\nabla u, \nabla \delta u)_\Omega \\ F(\delta u) &:= (f, \delta u)_\Omega, \end{aligned} \quad (2)$$

the variational problem of (1) reads:

$$\text{Find } u \in V := H_0^1(\Omega) : A(u, \delta u) = F(\delta u) \quad \forall \delta u \in V. \quad (3)$$

As in the FEM, the computational domain Ω is tessellated into smaller, non-overlapping elements E . Those are either polygons in 2D or polyhedrons in 3D. The tessellation will later be referred to as $\mathcal{T}_h$ and the edges and faces of the elements as e and $\mathcal{F}$. [The mesh size is \$h := \max_{E \in \mathcal{T}_h} h_E\$, while \$h_{\min}\$ denotes the smallest element diameter.](#) Based on the decomposition, we can define the discrete variational problem based on the finite dimensional subspace $V_h \subset V$ as

$$\text{Find } u_h \in V_h : A(u_h, \delta u_h) = F(\delta u_h) \quad \forall \delta u_h \in V_h. \quad (4)$$

Figure 1: Visualization of the basic idea of the projection u_h^π

1.1.1 VEM in 2D

We start the explanation by introducing the VEM for 2 dimensions. To anticipate the 3D implementation, where the planar faces of polyhedra behave as 2D elements, we denote the 2D element in this section as $\mathcal{F}$ (face).

The definition of the local function space V_1 reads

$$V_1(\mathcal{F}) := \left\{ v \in C^0(\bar{\mathcal{F}}) : v|_e \in \mathbb{P}_1(e) \text{ for any edge } e \subset \partial\mathcal{F}, \Delta v = 0 \right\}. \quad (5)$$

This means that the function space consists of all C^0 functions which are first-order polynomials on the boundary ($\mathbb{P}_1$ denotes the space of all first-order polynomials) with a Laplacian [equal to zero](#).

The global space V_h is a patchwork of these local spaces

$$V_h := \{ v_h \in H_0^1(\Omega) : v_h|_{\mathcal{F}} \in V_1(\mathcal{F}) \text{ for all } \mathcal{F} \in \mathcal{T}_h \}. \quad (6)$$

Consequently, if we know the values of the function $u_h \in V_h$ at all nodes, u_h is also known on all edges of the mesh (inside the elements, u_h is still not known).

Clearly, knowing u_h on the boundary is not enough to solve the given boundary value problem (1). As a remedy, the linear projection operator $\Pi_{1,\mathcal{F}}^\nabla$ is introduced (the subscript $\mathcal{F}$ will be omitted in the following). It needs to be computable, solely based on known values of u_h and it should give a "good" approximation of the true unknown function. The concept of the projection can be seen in Figure 1.

As depicted, the projected function $u_h^\pi = \Pi_1^\nabla u_h$ does not coincide with u_h on the

element boundary. It is rather defined to fulfill the two properties

$$(\nabla(u_h - u_h^\pi), \nabla p)_\mathcal{F} = 0 \quad \forall p \in \mathbb{P}_1 \quad (7a)$$

$$\frac{1}{N^{\text{dof}}} \sum_{i=1}^{N^{\text{dof}}} (\text{dof}_i(u_h) - \text{dof}_i(u_h^\pi)) = 0. \quad (7b)$$

Here, N^{dof} is the number of nodes per element and $\text{dof}_i(u_h)$ is the function u_h evaluated at the i -th degree of freedom of the element. As shown in [Ahmad et al. \(2013\)](#), for first and second order polynomials the derived projection operator additionally fulfills the property

$$(u_h - u_h^\pi, p)_\mathcal{F} = 0 \quad \forall p \in \mathbb{P}_1. \quad (8)$$

Utilizing the introduced projection, the bilinear form can be rewritten as

$$(\nabla u_h, \nabla \delta u_h)_\mathcal{F} = \underbrace{(\nabla u_h^\pi, \nabla \delta u_h^\pi)_\mathcal{F}}_{\text{consistency}} + \underbrace{(\nabla(u_h - u_h^\pi), \nabla(\delta u_h - \delta u_h^\pi))_\mathcal{F}}_{\text{stabilization}}. \quad (9)$$

The stabilization part cannot be computed exactly, as it requires the knowledge about u_h inside the element. In the remainder of this work, we choose as approximation of the stabilization term the so called "DOFI-DOFI" stabilization (cf. [Beirão Da Veiga et al. \(2014\)](#))

$$S_\mathcal{F}(u_h, \delta u_h) := \gamma_\mathcal{F} \sum_{i=1}^{N^{\text{dof}}} \text{dof}_i(u_h - u_h^\pi) \text{dof}_i(\delta u_h - \delta u_h^\pi). \quad (10)$$

1.1.2 VEM in 3D

The definition of the VEM in 3D is quite similar to the 2D case. Only the [definition](#) of the local space $V_1(\mathcal{V})$ on the polyhedron $\mathcal{V}$ differs

$$V_1(\mathcal{V}) := \{v \in H^1(\mathcal{V}) : v|_\mathcal{F} \in V_1(\mathcal{F}), \Delta v = 0\}. \quad (11)$$

The projection operator $\Pi_{1,\mathcal{V}}^\nabla$ needs to fulfill the same conditions as in the two dimensional case (7). But, in order to compute the projection operator for the volume, first the projection for all the volume faces needs to be computed.

1.1.3 VEM for linear elasticity

In the following section, we explain the application of the VEM for linear elasticity. To keep things general, E denotes an element in N_D dimensions with $N_D \in \{2, 3\}$. Let $\mathbf{V} \subset [H^1(\Omega)]^{N_D}$ denote the displacement space satisfying the homogeneous Dirichlet conditions, and let $\mathbf{V}_h \subset \mathbf{V}$ be its conforming VEM subspace. The discrete variational formulation reads:

$$\text{Find } \mathbf{u}_h \in \mathbf{V}_h : (\delta \boldsymbol{\varepsilon}_h, \mathbb{C}_0 : \boldsymbol{\varepsilon}_h)_\Omega - (\mathbf{t}, \delta \mathbf{u}_h)_{\partial\Omega_N} = 0 \quad \forall \delta \mathbf{u}_h \in \mathbf{V}_h. \quad (12)$$

Here, $\boldsymbol{\varepsilon}_h := \frac{1}{2}(\nabla \mathbf{u}_h + (\nabla \mathbf{u}_h)^T)$ denotes the strain of $\mathbf{u}_h$, and $\boldsymbol{\varepsilon}_h^\pi$ denotes the strain of $\mathbf{u}_h^\pi$. The tensor $\mathbb{C}_0$ is the elastic stiffness of the fully dense material, and $\mathbf{t}$ is the traction vector on the Neumann boundary $\partial\Omega_N$.

Writing the problem formulation as sum over all elements E and utilizing the VEM projection operator $\Pi_{1,E}^\nabla$ yields

$$\begin{aligned} \sum_{E \in \mathcal{T}_h} & \left[(\delta \boldsymbol{\varepsilon}_h^\pi, \mathbb{C}_0 : \boldsymbol{\varepsilon}_h^\pi)_E \right. \\ & + S_E(\mathbf{u}_h, \delta \mathbf{u}_h) \\ & \left. - (\mathbf{t}, \Pi_{1,\partial E}^\nabla \delta \mathbf{u}_h)_{\partial E \cap \partial\Omega_N} \right] = 0 \quad \forall \delta \mathbf{u}_h \in \mathbf{V}_h. \end{aligned} \quad (13)$$

where ∂E denotes the element faces in 3D and the element edges in 2D. The stabilization parameter γ_E is chosen as $\gamma_E = \frac{1}{4}\text{tr}(\mathbb{C}_0)$ in 2D and as $\gamma_E = \frac{h_E}{4}\text{tr}(\mathbb{C}_0)$ in 3D with h_E being the element diameter.

In the following, we use the global shorthand definitions

$$\begin{aligned} A(\mathbf{v}, \mathbf{w}) &:= \sum_{E \in \mathcal{T}_h} A_E(\mathbf{v}|_E, \mathbf{w}|_E), \\ S(\mathbf{v}, \mathbf{w}) &:= \sum_{E \in \mathcal{T}_h} S_E(\mathbf{v}|_E, \mathbf{w}|_E), \\ A_h(\mathbf{v}_h, \mathbf{w}_h) &:= A(\mathbf{v}_h^\pi, \mathbf{w}_h^\pi) + S(\mathbf{v}_h, \mathbf{w}_h). \end{aligned}$$

Here, $A_E(\mathbf{v}, \mathbf{w}) := (\boldsymbol{\varepsilon}(\mathbf{v}), \mathbb{C}_0 : \boldsymbol{\varepsilon}(\mathbf{w}))_E$ is the element elasticity bilinear form, and S_E denotes the element stabilization introduced above as $S_{\mathcal{F}}$ in two dimensions, including the projection of its arguments onto the kernel of $\Pi_{1,E}^\nabla$. Projections and derivatives are evaluated elementwise, so these expressions also apply to the generally discontinuous projected fields and their remainders.

1.2 Goal-oriented error estimation

In this work we employ a simplified version of the goal-oriented dual-weighted residual error estimation (DWR) method (cf. [Becker and Rannacher \(2001\)](#); [Bangerth and Rannacher \(2003\)](#)) to estimate element-wise error contributions w.r.t. a defined quantity of interest. However, we first introduce the main ideas of the standard DWR method, as it serves later as baseline to evaluate its simplified counterpart which is introduced in Section ???. We will limit ourselves to linear partial differential equations, while the framework itself can be extended to nonlinear equations as well. For a more detailed explanation of the DWR method we refer the reader to the exhaustive review book chapter [Endtmayer et al. \(2024\)](#) and our recent work [Sellmann and Junker \(2025\)](#) where we introduced a DWR framework for the Virtual Element Method.

The aim is to estimate a quantity of interest (QoI) which is most often some technical quantity from engineering, for instance stress or displacement evaluations. This QoI is defined by a functional $J : \mathbf{V} \rightarrow \mathbb{R}$ which can have an almost arbitrary form. Examples are:

$$\bullet J(\mathbf{u}) := \int_{\Omega} \mathbf{u}[1] + \mathbf{u}[2] \, d\Omega, \quad (14a)$$

$$\bullet J(\mathbf{u}) := \frac{1}{2} \int_{\Omega} \boldsymbol{\varepsilon} : \mathbb{C}_0 : \boldsymbol{\varepsilon} \, d\Omega, \quad (14b)$$

but classical norms can be presented by $J(\mathbf{u})$ as well. Here, the notation $\mathbf{u}[1] + \mathbf{u}[2]$ refers to the sum of the first two components of $\mathbf{u}$. Using a posteriori error analysis with DWR, errors measured in $J(\mathbf{u})$ can be estimated. These can be discretization, model or iteration errors. We denote the discrete goal functional by $J_h : \mathbf{V}_h \rightarrow \mathbb{R}$. For the strain energy in (14b), we use

$$J_h(\mathbf{v}_h) := \frac{1}{2} A_h(\mathbf{v}_h, \mathbf{v}_h).$$

For a Galerkin discretization, the error between the true and discrete goal values can be written as

$$\begin{aligned} J(\mathbf{u}) - J_h(\mathbf{u}_h) &= \frac{1}{2} \left(F(\mathbf{e}^*) - A(\mathbf{u}_h, \mathbf{e}^*) \right) \\ &\quad + \frac{1}{2} \left(D_{\mathbf{u}} J(\mathbf{u}_h)[\mathbf{e}] - A(\mathbf{e}, \mathbf{z}_h) \right) \\ &\quad + J(\mathbf{u}_h) - J_h(\mathbf{u}_h) + \mathcal{O}^3(\mathbf{e}, \mathbf{e}^*) \\ &= F(\mathbf{e}^*) - A(\mathbf{u}_h, \mathbf{e}^*) \end{aligned} \quad (15a)$$

$$+ J(\mathbf{u}_h) - J_h(\mathbf{u}_h) + \mathcal{O}^2(\mathbf{e}, \mathbf{e}^*). \quad (15b)$$

where we used the error $\mathbf{e} = \mathbf{u} - \mathbf{u}_h$, the adjoint error $\mathbf{e}^* = \mathbf{z} - \mathbf{z}_h$ and $D_{\mathbf{u}}J(\mathbf{u}_h)[\mathbf{e}]$ which marks the Gâteaux differential w.r.t $\mathbf{u}$. The remainder term is denoted as $\mathcal{O}$ and the superscript indicates its order of convergence with respect to the errors $\mathbf{e}$ and $\mathbf{e}^*$. The function $\mathbf{z}$ is the adjoint solution which can be interpreted as a sensitivity measure, weighting the element-wise error contributions with respect to the chosen goal functional. The adjoint solution is given by the problem:

$$\text{Find } \mathbf{z} \in \mathbf{V} : A(\delta \mathbf{u}, \mathbf{z}) = D_{\mathbf{u}}J(\mathbf{u})[\delta \mathbf{u}] \quad \forall \delta \mathbf{u} \in \mathbf{V}. \quad (16)$$

In this work, we utilize a quadratic goal functional and therefore employ the identity (15a) as the remainder term vanishes in this case.

As stated in the section before, the discrete solutions $\mathbf{z}_h$ and $\mathbf{u}_h$ are not explicitly known in the context of the VEM. We therefore approximate those by performing a triangulation of each element. For this, an additional node at the barycenter of each element is added. In consequence, each triangle consists of two adjacent boundary nodes of the element and the added barycenter node.

Remark 1. *A naive triangulation of the element using only the element nodes (without the additional node at the barycenter) yields unsatisfactory results (see Sellmann and Junker (2025)).*

The function value at the added node is calculated as

$$\mathbf{u}_0 = \mathbf{u}_0^\pi + \frac{1}{|\mathbf{E}|} \sum_{i=1}^{N^V} (\mathbf{u}_i^\pi + \mathbf{u}_{i+1}^\pi - \mathbf{u}_i - \mathbf{u}_{i+1}) |\mathbf{E}_i|, \quad (17)$$

where N^V denotes the number of nodes of the element and $|\mathbf{E}_i|$ is the area of the i -th triangle built by the nodes $i, i+1$ and the barycenter node 0. Then, we approximate $\mathbf{u}_h$ for each triangle as a linear polynomial matching the function values at the three triangle nodes. By this we ensure that the property $\int_E \mathbf{u}_h \, dE = \int_E \mathbf{u}_h^\pi \, dE$ holds.

In Sellmann and Junker (2025) it has been shown that the stabilization error needs to be incorporated in the error identities (15) since the derivation of the DWR error estimator relies on Galerkin orthogonality between the true and approximated solution, which does not hold for the VEM approximations. For this, we introduce the bilinear correction form

$$S^{\text{corr}}(\mathbf{v}_h, \mathbf{w}_h) := A(\mathbf{v}_h - \mathbf{v}_h^\pi, \mathbf{w}_h - \mathbf{w}_h^\pi) - S(\mathbf{v}_h, \mathbf{w}_h).$$

The adapted error identities for $J(\mathbf{u}) - J_h(\mathbf{u}_h)$ read

$$\begin{aligned} J(\mathbf{u}) - J_h(\mathbf{u}_h) &= \frac{1}{2} \left(F(\mathbf{e}^*) - A(\mathbf{u}_h, \mathbf{e}^*) \right) \\ &\quad + \frac{1}{2} \left(D_{\mathbf{u}} J(\mathbf{u}_h)[\mathbf{e}] - A(\mathbf{e}, \mathbf{z}_h) \right) \\ &\quad - S^{\text{corr}}(\mathbf{u}_h, \mathbf{z}_h) + J(\mathbf{u}_h) - J_h(\mathbf{u}_h) + \mathcal{O}^3(\mathbf{e}, \mathbf{e}^*) \end{aligned} \quad (18a)$$

$$\begin{aligned} &= F(\mathbf{e}^*) - A(\mathbf{u}_h, \mathbf{e}^*) - S^{\text{corr}}(\mathbf{u}_h, \mathbf{z}_h) \\ &\quad + J(\mathbf{u}_h) - J_h(\mathbf{u}_h) + \mathcal{O}^2(\mathbf{e}, \mathbf{e}^*). \end{aligned} \quad (18b)$$

Since the functions $\mathbf{u}$ and $\mathbf{z}$ are unknown in most practical applications they need to be approximated. In this work we utilize three different techniques. Straightforward approaches are e.g. the computation of these functions on finer meshes or the utilization of higher order ansatz spaces. As baseline, we employ $\mathbf{z} \approx \mathbf{z}_h^{(2)}$ as it is done e.g. in Richter and Wick (2015); Endtmayer et al. (2024). As the computation of higher-order functions is quite expensive, different reconstruction techniques have been proposed to obtain a higher order approximation in a post-processing step. However, most of them fall short for general polygonal elements and/or if the shape functions are not known everywhere (cf. Becker and Rannacher (2001); Rognes and Logg (2012); Carpio et al. (2013)). For polygonal meshes Weißer and Wick (2018) proposed a method where each geometrical edge needs to consist of an even number of nodes. If this is not the case, additional unknowns are added to the global problem which can increase the computational cost dramatically. Therefore, we utilize the **G**auss-**P**oint **R**econstruction **M**ethod with **s**tatic **c**ondensation (**GPRMSC**) as reconstruction technique which has been proposed in our work Sellmann and Junker (2025).

The third method simply uses $\mathbf{z} \approx \mathbf{z}_h^\pi$. For more details on the latter two methods see Section ??.

1.2.1 Error localization

The error estimator presented so far does only allow for an estimation of the global error. Since we want to utilize the error estimator to indicate which elements should be refined, local error contributions are required. To do so, we utilize the partition of unity (PU) based localization strategy presented in Richter and Wick (2015). For this we use the fact that the sum of all projected shape functions always equals a vector filled with ones at every location in the domain Ω , i.e.,

$$\mathbf{1} \equiv \sum_i^m \boldsymbol{\varphi}_i^\pi(\mathbf{x}) \quad \forall \mathbf{x} \in \Omega. \quad (19)$$

Here, m denotes the total number of all shape functions. The FEM and VEM constitute by construction PUs and therefore, this idea can easily be implemented in existing codes. Afterward, we multiply the adjoint terms point-wise (denoted by $\odot$) by the projected shape functions. For the strain-energy goal, the self-adjoint identity (??) yields

$$J(\mathbf{u}) - J_h(\mathbf{u}_h) \approx \frac{1}{2} \sum_i^m \left[F(\mathbf{e}^* \odot \boldsymbol{\varphi}_i^\pi) - A(\mathbf{u}_h, \mathbf{e}^* \odot \boldsymbol{\varphi}_i^\pi) - S_i^{\text{corr}}(\mathbf{u}_h, \mathbf{z}_h) \right] = \sum_i^m \eta_i. \quad (20)$$

Here, S_i^{corr} denotes the PU contribution of the correction term, with $\sum_i^m S_i^{\text{corr}} = S^{\text{corr}}$. As a result, each η_i represents the error associated with the i -th degree of freedom. The element-wise error can be calculated afterward by adding the weighted absolute error contributions of all degrees of freedom associated with the element. The weights are determined by the number of elements containing a degree of freedom. E.g. if 5 elements contain a degree of freedom its weight is $\frac{1}{5}$. In the remainder of this paper, we utilize two metrics to measure the quality of the derived error estimator. The first metric is the effectivity index, defined as

$$\text{idx}_{\text{eff}} := \left| \frac{\sum_i^m \eta_i}{J(\mathbf{u}) - J_h(\mathbf{u}_h)} \right|. \quad (21)$$

A value of one indicates that the global error is estimated perfectly. The second metric measures the quality of the error localization by summing the absolute local error contributions, which are later used for the mesh refinement process

$$\text{idx}_{\text{ind}} := \frac{\sum_i^m |\eta_i|}{|J(\mathbf{u}) - J_h(\mathbf{u}_h)|}. \quad (22)$$

Again, a value of one is optimal. If the value exceeds 1, it indicates that the sum of local error contributions exceeds the global error (known as overestimation), potentially leading to the refinement of more elements than necessary.

1.3 Thermodynamic Topology Optimization

In the following section, we briefly recall the foundations of the thermodynamic topology optimization (TTO). For more details, we refer the reader to [Jantos et al. \(2019\)](#) and [Junker and Balzani \(2021b\)](#).

The final TTO algorithm is broadly comparable to standard topology optimization

approaches, but with two main differences: (1) the filter is directly integrated into the Hamilton functional which leads to a well posed problem, and (2) there is no need for an external optimization algorithm since the density update is modeled as an evolution equation over time. These features enable a straightforward integration into existing FEM software.

If strictly derived as an inverse damage model, TTO is limited to self-adjoint problems. However, the governing equations can be derived differently, broadening the space of possible applications without losing the advantages listed above.

Here, the underlying functional is closely related to the Lagrangian used to derive the equations for the DWR framework described in the previous chapter:

$$\mathcal{L}(\mathbf{u}, \mathbf{z}, \chi) = J(\mathbf{u}, \chi) - D_{\mathbf{u}}\mathcal{G}(\mathbf{u}, \chi)[\mathbf{z}] + \mathcal{D}(\chi) + \mathcal{F}(\chi) + \mathcal{C}(\chi). \quad (23)$$

The functional $\mathcal{G} := (\Psi, 1)_{\Omega} - (\mathbf{f}, \mathbf{u})_{\Omega} - (\mathbf{t}, \mathbf{u})_{\partial\Omega}$ denotes the Gibbs energy, where the Helmholtz free energy Ψ takes the form $\Psi = \frac{1}{2}(\chi^3 \nabla \mathbf{u}, \mathbb{C}_0 : \nabla \mathbf{u})$ for linear elasticity.

The variable $\chi(\mathbf{x}) \in [\chi_{\min}, 1]$ serves as a continuous design indicator, where $\chi = 1$ signifies full material and $\chi = \chi_{\min}$ represents the absence of material.

Following the SIMP (Solid Isotropic Material with Penalization) approach, the stiffness tensor is defined as

$$\mathbb{C}(\chi) = \chi^p \mathbb{C}_0,$$

where $p = 3$ is the penalization exponent used in this work. This penalization drives the solution towards either solid or void states within the design domain Ω . The functionals $\mathcal{D}$, $\mathcal{F}$, and $\mathcal{C}$ determine the behavior of the optimization algorithm. Here,

$$\mathcal{D} := (p_{\text{diss}}^*, \dot{\chi})_{\Omega}$$

is the dissipation functional with the positive viscosity η . The dissipation force p_{diss} is derived from a dissipation functional $p_{\text{diss}} := \frac{\partial \Delta^{\text{diss}}}{\partial \dot{\chi}}$ as is common in the field of material theory. This is in accordance with the work Junker and Balzani (2021a). The asterisk indicates that the dissipative force is modeled and therefore the stationarity w.r.t this quantity is not required. In this work we define $\Delta^{\text{diss}} := \frac{\eta}{2} \dot{\chi}^2$. The regularization function $\mathcal{F}$ is defined as

$$\mathcal{F} := \frac{1}{2}(\beta \|\nabla \chi\|, \|\nabla \chi\|)_{\Omega},$$

where the parameter $\beta \in \mathbb{R}^+$ controls the minimum member size to ensure a checkerboard-free structure.

Finally, the constraint functional is defined as

$$\mathcal{C} = (\chi - \varrho_0, \lambda)_\Omega + (\gamma, \chi)_\Omega.$$

The first term ensures the conservation of the total structural mass throughout the optimization process via the Lagrange multiplier $\lambda \in \mathbb{R}$. The second term ensures that the design variable χ remains within the interval $[\chi_{\min}, 1]$ using the Kuhn-Tucker parameter γ , which is defined as:

$$\gamma = \begin{cases} \bar{\gamma} & : (\dot{\chi} > 0) \wedge (\chi = 1) \\ -\bar{\gamma} & : (\dot{\chi} < 0) \wedge (\chi = \chi_{\min}) \\ 0 & : \text{else} \end{cases} \quad (24)$$

Afterward, the stationary condition of $\mathcal{L}$ w.r.t $\{\mathbf{u}, \mathbf{z}, \chi\}$ yields all the governing equations required for the optimization procedure. First, the stationarity condition w.r.t $\mathbf{z}$ yields the well known balance of momentum

$$D_{\mathbf{u}}\mathcal{G}(\mathbf{u}, \chi)[\delta\mathbf{z}] = 0 \quad \forall \delta\mathbf{z} \in \mathbf{V}. \quad (25)$$

Variation w.r.t $\mathbf{u}$ yields the governing equation for the adjoint function $\mathbf{z}$

$$D_{\mathbf{u}}J(\mathbf{u}, \chi)[\delta\mathbf{u}] - D_{\mathbf{uu}}\mathcal{G}(\mathbf{u}, \chi)[\mathbf{z}, \delta\mathbf{u}] = 0 \quad \forall \delta\mathbf{u} \in \mathbf{V}. \quad (26)$$

And finally, the variation w.r.t the design variable χ yields

$$\begin{aligned} & D_{\chi}J(\mathbf{u}, \chi)[\delta\chi] - D_{\mathbf{u},\chi}\mathcal{G}(\mathbf{u}, \chi)[\mathbf{z}, \delta\chi] + (\eta\dot{\chi}, \delta\chi)_\Omega \\ & + (\beta\nabla\chi, \nabla\delta\chi)_\Omega + (\lambda, \delta\chi)_\Omega + (\gamma, \delta\chi)_\Omega = 0 \quad \forall \delta\chi \in H^1(\Omega). \end{aligned} \quad (27)$$

In this work, we limit our scope to strain energy minimization with the goal functional $J(\mathbf{u}, \chi) = \frac{1}{2}(\chi^3\boldsymbol{\varepsilon}, \mathbb{C}_0 : \boldsymbol{\varepsilon})$ for linear elastic materials. Under those assumptions the problem is self adjoint, and we recover the standard governing equations for the TTO as they are derived in Junker and Balzani (2021b). The equation for $\mathbf{z}$ simplifies to

$$(\chi^3\nabla\mathbf{u}, \mathbb{C}_0 : \nabla\delta\mathbf{u}) = (\chi^3\nabla\mathbf{z}, \mathbb{C}_0 : \nabla\delta\mathbf{u}) \quad \forall \delta\mathbf{u} \in \mathbf{V} \quad (28)$$

which directly yields $\mathbf{z} = \mathbf{u}$. However, we wish to highlight that the adjoint required for the DWR framework is the same as that required for the present optimization approach. Consequently, even for non-self-adjoint problems, the added computational cost is limited, which strongly motivates the use of the DWR

method with TTO. Further, the evolution equation for χ transforms into

$$-(\frac{\partial \Psi}{\partial \chi}, \delta \chi)_\Omega + (\lambda + \gamma, \delta \chi)_\Omega + (\eta \dot{\chi}, \delta \chi)_\Omega + (\beta \nabla \chi, \nabla \delta \chi)_\Omega \ni 0 \quad \forall \delta \chi \in H^1(\Omega). \quad (29)$$

The latter is transformed into its strong form, yielding purely local update equations

$$\begin{aligned} p_\chi + \eta \dot{\chi} + \lambda + \gamma - \nabla \cdot (\beta \nabla \chi) &\ni 0 & \forall \mathbf{x} \in \Omega \\ \beta \nabla \chi \cdot \mathbf{n} &= 0 & \forall \mathbf{x} \in \partial \Omega \end{aligned} \quad (30)$$

with the thermodynamic driving force

$$p_\chi = -\frac{p}{2} \chi^{p-1} \boldsymbol{\varepsilon} : \mathbb{C}_0 : \boldsymbol{\varepsilon}. \quad (31)$$

1.3.1 Numerical treatment of the density update

The optimization is performed in a staggered manner. First, the displacement field is computed for the current density field. Afterward, the density is updated using the evolution equation (30). The update of the density variable is based on the neighbored element method (NEM), which provides a highly efficient numerical scheme. The NEM can be seen as a generalization of the finite difference method to non-regular meshes (see Jantos et al. (2019); Junker et al. (2019); Vogel and Junker (2020); Junker and Balzani (2021b)). In the following, the algorithmic details of the scheme are explained.

First, the evolution equation (30) is discretized using an explicit Euler scheme:

$$\chi^{i+1} = \chi^i + \frac{\Delta t}{\eta} (-p_\chi^i - \lambda - \gamma + \nabla \cdot (\beta \nabla \chi^i)). \quad (32)$$

Here, i denotes the density-update step. Following Jantos et al. (2019), the quantities η and β are computed from the user-defined regularization parameter β^* , which controls the minimum structure size, and the viscosity parameter η^* . Their default values are $2h_{\min}^2$ and 15, respectively.

For the numerical update, we use the positive element-averaged driving force $p_E := -|\mathbf{E}|^{-1} \int_E p_\chi dV$. The average driving force is defined as

$$p_{\text{avg}} := \frac{\int_\Omega p_E g(\chi) dV}{\int_\Omega g(\chi) dV}, \quad g(\chi) := (\chi - \chi_{\min})(\chi_{\max} - \chi). \quad (33)$$

The viscosity and element-wise regularization parameter are then computed as

$$\eta = p_{\text{avg}}\eta^*, \quad \beta_E = \max \{p_{\text{avg}}\beta^*, p_{\text{avg}}h_E^2\}, \quad (34)$$

where h_E denotes the element diameter. The use of the maximum has been proposed in [Vogel and Junker \(2021\)](#) to account for large differences in element size arising from adaptive mesh refinement.

For the density update, the regularization term $\nabla \cdot (\beta \nabla \chi)$ needs to be computed. To this end, the procedure proposed in [Jantos et al. \(2019\)](#), with the modifications suggested in [Cihan et al. \(2024\)](#), is employed and extended to account for spatial variations of β . The element-wise values of χ and β are interpreted as samples of underlying sufficiently smooth fields at the element barycentres. Their derivatives are reconstructed from neighbouring samples using local Taylor expansions. For completeness, the underlying procedure is described below. The displayed equations apply to the two-dimensional case and can be extended analogously to three dimensions.

Since the evolution equation (30) is evaluated at the element barycentres, the required derivatives are reconstructed at the same locations. A Taylor expansion of the design variable χ about the barycentre $\mathbf{x}_j$ of element j gives

$$\begin{aligned} \chi(\mathbf{x}_i) = & \chi(\mathbf{x}_j) + \frac{\partial \chi(\mathbf{x}_j)}{\partial x} \Delta x_i + \frac{\partial \chi(\mathbf{x}_j)}{\partial y} \Delta y_i \\ & + \frac{1}{2} \left[\frac{\partial^2 \chi(\mathbf{x}_j)}{\partial x^2} \Delta x_i^2 + 2 \frac{\partial^2 \chi(\mathbf{x}_j)}{\partial x \partial y} \Delta x_i \Delta y_i + \frac{\partial^2 \chi(\mathbf{x}_j)}{\partial y^2} \Delta y_i^2 \right] + \mathcal{O}(\|\mathbf{x}_i - \mathbf{x}_j\|^3). \end{aligned} \quad (35)$$

Here, $\Delta x_i = x_i - x_j$ and $\Delta y_i = y_i - y_j$. Collecting the expansions for all elements sharing a node with element j yields the local least-squares system

$$\underbrace{\begin{bmatrix} \frac{1}{2}\Delta x_1^2 & \frac{1}{2}\Delta y_1^2 & \Delta x_1\Delta y_1 & \Delta x_1 & \Delta y_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{1}{2}\Delta x_n^2 & \frac{1}{2}\Delta y_n^2 & \Delta x_n\Delta y_n & \Delta x_n & \Delta y_n \end{bmatrix}}_{\mathbf{G}} \underbrace{\begin{bmatrix} \chi_{,xx} \\ \chi_{,yy} \\ \chi_{,xy} \\ \chi_{,x} \\ \chi_{,y} \end{bmatrix}}_{\mathbf{m}} \approx \underbrace{\begin{bmatrix} \chi_1 - \chi_j \\ \vdots \\ \chi_n - \chi_j \end{bmatrix}}_{\mathbf{b}}. \quad (36)$$

In three dimensions, the neighbourhood consists of elements sharing an edge, and the Taylor expansion includes the additional first and second derivatives involving the third coordinate.

The derivative vector $\mathbf{m}$ is approximated by the regularized least-squares solution

$$\mathbf{m} = (\mathbf{G}^T \mathbf{G} + \alpha \mathbf{I})^{-1} \mathbf{G}^T \mathbf{b}. \quad (37)$$

The term $\alpha \mathbf{I}$ stabilizes the reconstruction when the distribution of neighbouring points leads to a singular or ill-conditioned matrix $\mathbf{G}^T \mathbf{G}$. The reconstructed Laplacian and first derivatives are

$$\nabla^2 \chi_j \approx \mathbf{m}[1] + \mathbf{m}[2], \quad \chi_{,x,j} \approx \mathbf{m}[4], \quad \chi_{,y,j} \approx \mathbf{m}[5]. \quad (38)$$

For spatially varying β , the product rule gives

$$\nabla \cdot (\beta \nabla \chi) = \beta \nabla^2 \chi + \beta_{,x} \chi_{,x} + \beta_{,y} \chi_{,y}. \quad (39)$$

The derivatives of β are reconstructed using the same least-squares procedure, replacing $\chi_i - \chi_j$ by $\beta_i - \beta_j$ in the right-hand side of (36). Thus, no additional geometric reconstruction is required. The regularization term at $\mathbf{x}_j$ is approximated as

$$[\nabla \cdot (\beta \nabla \chi)]_j \approx \beta_j (\mathbf{m}[1] + \mathbf{m}[2]) + \beta_{,x,j} \mathbf{m}[4] + \beta_{,y,j} \mathbf{m}[5], \quad (40)$$

where $\beta_{,x,j}$ and $\beta_{,y,j}$ denote the reconstructed coefficient derivatives. For constant β , the additional terms vanish and the original Laplacian-based expression is recovered.

The derivative reconstructions can be assembled for all elements as sparse matrices:

$$\nabla^2 \chi \approx \mathbf{A} \chi, \quad \chi_{,x} \approx \mathbf{A}_x \chi, \quad \chi_{,y} \approx \mathbf{A}_y \chi, \quad (41)$$

with the approximations understood at the element barycentres. Consequently, the complete regularization term can be evaluated as

$$\mathbf{r} := \mathbf{R}(\beta) \chi \approx [\nabla \cdot (\beta \nabla \chi)]_h, \quad (42)$$

where

$$\mathbf{R}(\beta) = \text{diag}(\beta) \mathbf{A} + \text{diag}(\mathbf{A}_x \beta) \mathbf{A}_x + \text{diag}(\mathbf{A}_y \beta) \mathbf{A}_y. \quad (43)$$

Here, $\text{diag}(\mathbf{v})$ denotes the diagonal matrix with diagonal entries given by $\mathbf{v}$. The matrices $\mathbf{A}$, $\mathbf{A}_x$, and $\mathbf{A}_y$ depend only on the sampling geometry. They can therefore be computed once for a fixed mesh and are recomputed after each remeshing. Changes in β require only an update of the coefficient-dependent terms in (43), without repeating the local least-squares reconstruction.

Boundary elements

For boundary elements, the homogeneous Neumann condition $\beta \nabla \chi \cdot \mathbf{n} = 0$ must be enforced. Since $\beta > 0$, this is equivalent to $\nabla \chi \cdot \mathbf{n} = 0$. To this end, a ghost state is introduced for each boundary edge, or boundary face in three dimensions. For an element with barycentre $\mathbf{x}_j$, its position is

$$\mathbf{x}_{\text{ghost}} = \mathbf{x}_j + 2(\mathbf{p} \cdot \mathbf{n})\mathbf{n}, \quad (44)$$

where $\mathbf{n}$ is the outward unit normal and $\mathbf{p}$ connects the barycentre to an arbitrary point on the associated boundary edge or face; see Figure 2. The ghost density is set to $\chi_{\text{ghost}} = \chi_j$, representing a vanishing normal derivative. For the coefficient reconstruction, the same reflected sampling geometry is used with $\beta_{\text{ghost}} = \beta_j$.

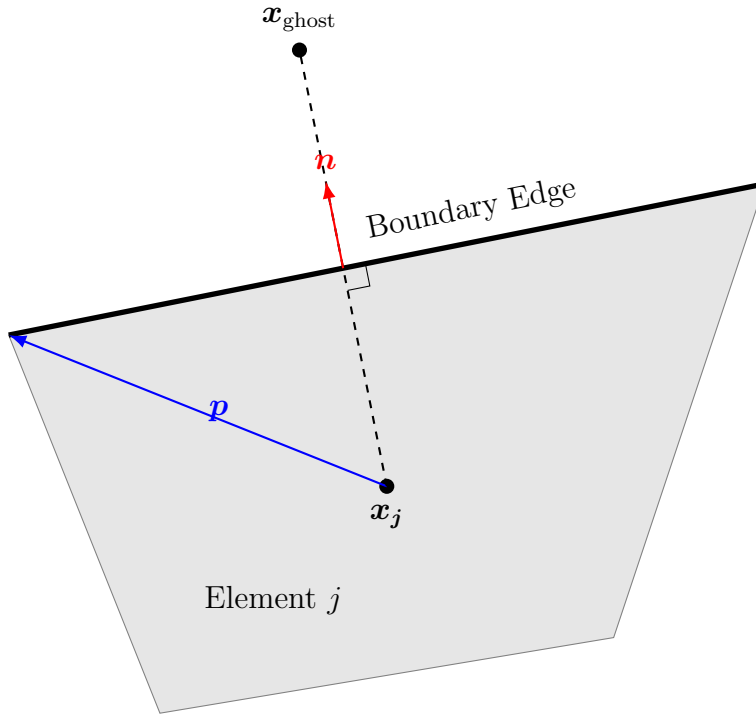


Figure 2: Computation of the ghost state position

The complete density update is summarized in Algorithm 1. The density bounds are enforced by projection onto $[\chi_{\min}, \chi_{\max}]$, while the Lagrange multiplier λ is determined by bisection to satisfy the prescribed volume fraction. The bisection interval accounts for both the driving force and the regularization contribution.

Algorithm 1 Density update.

```

1: Given:  $\chi_{\min}, \chi_{\max}, \eta^*, \beta^*, \Delta t^*, \varrho_0, h_{\min}, \epsilon_{\text{density}}$ 
2: Input:  $\mathbf{u}, \chi$ 
3: On grid change: compute  $\mathbf{A}, \mathbf{A}_x, \mathbf{A}_y$  from (36), including the ghost states
4:  $n_{\text{steps}} := \max \left\{ 1, \left\lceil \frac{12}{\eta^*} \frac{\beta^*}{h_{\min}^2} \right\rceil \right\}, \quad \Delta t := \Delta t^* / n_{\text{steps}}$ 
5: for  $i = 1, \dots, n_{\text{steps}}$  do
6:   Compute  $p_E := \frac{1}{|E|} \int_E \frac{3}{2} \chi^2 \boldsymbol{\varepsilon} : \mathbb{C}_0 : \boldsymbol{\varepsilon} \, dV$  for all  $E \in \mathcal{T}_h$ 
7:    $p_{\text{avg}} := \frac{\sum_{E \in \mathcal{T}_h} |E| p_E g(\chi_E)}{\sum_{E \in \mathcal{T}_h} |E| g(\chi_E)}, \quad g(\chi) := (\chi - \chi_{\min})(\chi_{\max} - \chi)$ 
8:    $\eta := p_{\text{avg}} \eta^*, \quad \beta_E := \max \{ p_{\text{avg}} \beta^*, p_{\text{avg}} h_E^2 \}$  for all  $E \in \mathcal{T}_h$ 
9:   Compute  $\mathbf{r} := \mathbf{R}(\boldsymbol{\beta}) \boldsymbol{\chi}$  using (43)
10:   $s_E := p_E + r_E$  for all  $E \in \mathcal{T}_h$ 
11:   $s_{\min} := \min_{E \in \mathcal{T}_h} s_E, \quad s_{\max} := \max_{E \in \mathcal{T}_h} s_E$ 
12:   $\lambda_{\min} := s_{\min} - \frac{\eta}{\Delta t} (\chi_{\max} - \chi_{\min}), \quad \lambda_{\max} := s_{\max} + \frac{\eta}{\Delta t} (\chi_{\max} - \chi_{\min})$ 
13:   $\lambda := (\lambda_{\min} + \lambda_{\max}) / 2$ 
14:  for  $n_{\text{bisect}} = 1, 2, \dots$  do
15:    for  $E \in \mathcal{T}_h$  do
16:       $\chi_E^{\text{tr}} := \chi_E + \frac{\Delta t}{\eta} (p_E - \lambda + r_E)$ 
17:       $\chi_E^{\text{tr}} := \max \{ \chi_{\min}, \min \{ \chi_{\max}, \chi_E^{\text{tr}} \} \}$ 
18:    end for
19:     $\rho^{\text{tr}} := \frac{1}{|\Omega|} \sum_{E \in \mathcal{T}_h} |E| \chi_E^{\text{tr}}$ 
20:    if  $|\rho^{\text{tr}} - \varrho_0| < \epsilon_{\text{density}}$  then
21:      break
22:    end if
23:    if  $\rho^{\text{tr}} > \varrho_0$  then
24:       $\lambda_{\min} := \lambda$ 
25:    else
26:       $\lambda_{\max} := \lambda$ 
27:    end if
28:     $\lambda := (\lambda_{\min} + \lambda_{\max}) / 2$ 
29:  end for
30:   $\chi := \chi^{\text{tr}}$ 
31: end for

```

For the density update, the regularization term $\nabla \cdot (\beta \nabla \chi)$ needs to be computed. We approximate this term using a cell-centred finite-volume formulation with a

two-point flux approximation (TPFA), following (Droniou, 2014, Section 2). For spatially constant β , the standard Taylor-based TTO treatment approximates the density field locally by a Taylor expansion and reconstructs the Laplacian required to evaluate $\nabla \cdot (\beta \nabla \chi) = \beta \nabla^2 \chi$ Jantos et al. (2019). Here, we deviate from this treatment for two reasons. First, the element-size dependence in (34) can lead to large differences in β between adjacent elements, so the regularization term can no longer be reduced to $\beta \nabla^2 \chi$. The flux formulation accounts for these differences by treating β as a piecewise constant coefficient in the face transmissibilities. Second, each TPFA face flux requires only the two adjacent cell values, avoiding the wider neighbourhood search and local least-squares reconstruction of the Taylor approach. The reduced neighbourhood simplifies rebuilding the regularization operator on adaptively refined meshes, where collecting the larger Taylor stencils can be time-consuming.

Let $K \in \mathcal{T}_h$ denote a cell with volume $|K|$, barycentre $\mathbf{x}_K$, and density χ_K . Integrating (30) over K and dividing by $|K|$ gives

$$\eta \dot{\chi}_K - p_K + \lambda + \gamma_K - \bar{r}_K \ni 0, \quad \bar{r}_K := \frac{1}{|K|} \int_K \nabla \cdot (\beta \nabla \chi) \, dV. \quad (45)$$

Here, χ_K represents the cell average, $p_K = -|K|^{-1} \int_K p_\chi \, dV$ is the positive driving force defined above, and γ_K is the cell-wise bound reaction. In the discrete problem, the density and the strain obtained from the first-order VEM displacement projection are constant within each element. Consequently, the corresponding elastic driving force is also cell-wise constant. The viscosity η and the volume multiplier λ are spatially constant, while the bound constraint is imposed separately for each cell. Thus, averaging leaves these discrete quantities unchanged; only the regularization term requires information from neighbouring cells. For a higher-order displacement approximation, the nonlinear elastic driving force would instead have to be integrated over the cell.

Introducing the regularization flux $\mathbf{j} := -\beta \nabla \chi$, the divergence theorem yields

$$\bar{r}_K = -\frac{1}{|K|} \sum_{f \in \partial K} Q_{K,f}, \quad Q_{K,f} := \int_f \mathbf{j} \cdot \mathbf{n}_{K,f} \, dA = - \int_f \beta \nabla \chi \cdot \mathbf{n}_{K,f} \, dA. \quad (46)$$

The vector $\mathbf{n}_{K,f}$ is the outward unit normal of cell K on face f , consistent with the boundary-normal notation in (30). The quantity $Q_{K,f}$ is the integrated outward flux, so a positive value contributes to a decrease of χ_K .

For an interior face $f = K|L$, the normal points from K towards its neighbour L .

With $\mathbf{x}_f$ any point in the face plane, define the normal distances

$$\delta_{K,f} := (\mathbf{x}_f - \mathbf{x}_K) \cdot \mathbf{n}_{K,f}, \quad \delta_{L,f} := (\mathbf{x}_L - \mathbf{x}_f) \cdot \mathbf{n}_{K,f}. \quad (47)$$

Both distances are assumed positive, i.e., the cell centres lie on the respective interior sides of the common face. For $\beta_K, \beta_L > 0$, the TPFA flux is

$$Q_{K,f} \approx F_{K,f} := T_f(\chi_K - \chi_L), \quad T_f := \frac{|f|}{\delta_{K,f}/\beta_K + \delta_{L,f}/\beta_L}, \quad F_{L,f} = -F_{K,f}. \quad (48)$$

Here, $|f|$ is the face area, or the edge length in two dimensions. The denominator accounts for the two cell-to-face distances and their respective coefficients. It follows from eliminating a common face value between the two one-sided flux approximations while imposing continuity of the normal flux (Droniou, 2014, Equations (2.2)–(2.4)). At a coarse–fine interface, each active subface is treated separately with its two incident cells. On the exterior boundary, $F_{K,f} = 0$ directly enforces the homogeneous Neumann condition.

Substitution into (45) gives the semi-discrete density evolution

$$\eta \dot{\chi}_K - p_K + \lambda + \gamma_K - \frac{1}{|K|} \sum_{f=K|L} T_f(\chi_L - \chi_K) \ni 0, \quad (49)$$

where the sum runs over the interior faces of K . On an orthogonal mesh, the vector $\mathbf{x}_L - \mathbf{x}_K$ is parallel to $\mathbf{n}_{K,f}$. On a nonorthogonal mesh, the centre-to-centre density difference also contains a contribution from the tangential variation of χ . The TPFA formula omits the corresponding tangential correction and therefore does not, in general, provide a consistent face-flux approximation on such meshes. Its suitability for regularizing the density field on the nonorthogonal meshes considered here remains to be verified numerically.

Collecting the coefficients of the cell densities yields the sparse matrix-vector product

$$\mathbf{r}^{\text{TPFA}} = \mathbf{R}_{\text{TPFA}}(\boldsymbol{\beta})\boldsymbol{\chi}, \quad r_K^{\text{TPFA}} = \frac{1}{|K|} \sum_{f=K|L} T_f(\chi_L - \chi_K). \quad (50)$$

If $\mathcal{N}(K)$ denotes the set of face neighbours of K , the matrix entries are

$$(\mathbf{R}_{\text{TPFA}}(\boldsymbol{\beta}))_{KL} = \begin{cases} \frac{1}{|K|} \sum_{f=K|L} T_f, & L \in \mathcal{N}(K), \\ -\frac{1}{|K|} \sum_{f=K|J} T_f, & L = K, \\ 0, & \text{otherwise.} \end{cases} \quad (51)$$

Thus, each row contains only a diagonal entry and contributions from face neighbours. The row sums vanish, so a constant density field produces no regularization force.

Finally, the common driving-force factor in (34) can be separated from the geometric coefficients. For $p_{\text{avg}} > 0$, write

$$\hat{\beta}_K := \frac{\beta_K}{p_{\text{avg}}} = \max\{\beta^*, h_K^2\}, \quad \mathbf{R}_{\text{TPFA}}(\boldsymbol{\beta}) = p_{\text{avg}} \hat{\mathbf{R}}_{\text{TPFA}}, \quad (52)$$

where $\hat{\mathbf{R}}_{\text{TPFA}}$ is assembled using $\hat{\beta}_K$ in place of β_K . This normalized matrix is constant for a fixed mesh and fixed user parameters and is rebuilt after mesh adaptation. Using $\eta = p_{\text{avg}}\eta^*$, the explicit Euler update with bound projection becomes

$$\chi_K^{i+1} = \Pi_{[\chi_{\min}, \chi_{\max}]} \left[\chi_K^i + \frac{\Delta t}{\eta^*} \left(\frac{p_K^i - \lambda^i}{p_{\text{avg}}^i} + (\hat{\mathbf{R}}_{\text{TPFA}} \boldsymbol{\chi}^i)_K \right) \right]. \quad (53)$$

Here, $\Pi_{[\chi_{\min}, \chi_{\max}]}$ denotes clipping to the admissible density interval and implements the bound reaction γ_K . At each density-update step, the scalar multiplier λ^i is determined by bisection such that $\sum_{K \in \mathcal{T}_h} |K| \chi_K^{i+1} = \varrho_0 |\Omega|$.

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